

$\wedge$ = and

$\vee$ = or

100

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Quiz 7

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1. Write a truth table for

a) $(p \wedge q) \Rightarrow p$

b) $(p \Rightarrow q) \Leftrightarrow (\neg p \Rightarrow \neg q)$

c) $(\neg p \wedge (p \Rightarrow q)) \Rightarrow \neg q$

d) $(p \wedge r) \Rightarrow q$

a

p	q	$p \wedge q$	$p \wedge q \Rightarrow p$
F	F	F	T
F	T	F	T
T	F	F	T
T	T	T	T

b

p	q	$p \Rightarrow q$	$\neg p$	$\neg q$	$\neg p \Rightarrow \neg q$	Q
F	F	T	T	T	T	T
F	T	T	T	F	F	F
T	F	F	F	T	T	F
T	T	T	F	F	T	T

c

p	q	$\neg p$	$\neg q$	$p \Rightarrow q$	$\neg p \wedge (p \Rightarrow q)$	$Q \Rightarrow \neg q$
F	F	T	T	T	T	T
F	T	T	F	T	T	F
T	F	F	T	F	F	T
T	T	F	F	T	F	T

d

p	q	r	$p \wedge r$	$p \wedge r \Rightarrow q$
F	F	F	F	T
F	F	T	F	T
F	T	F	F	T
F	T	T	F	T
T	F	F	F	T
T	F	T	T	F
T	T	F	F	T
T	T	T	T	T

2. Which from #1 are tautologies?

a